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An experimental and kinetic modeling study of three butene isomers pyrolysis at low pressure

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ABSTRACT

Pyrolysis of three butene isomers (C_4H_8) including 1-butene (1-C_4H_8), 2-butene (2-C_4H_8) and *i*-butene (*i*C_4H_8) were studied from 900 to 1900 K at low pressure. Synchrotron vacuum ultraviolet (VUV) photoionization mass spectrometry with molecular-beam sampling technique was used for isomeric identification of products and intermediates and also for concentration measurement. Based on the experimental results, a kinetic model consisting of 76 species and 232 reactions was developed to simulate mole fractions of species. The mole fraction profiles of pyrolysis species predicted by the model are in good agreement with the experimental measurements. The decomposition pathways of C_4H_8 are illustrated according to the reaction flux analysis. Our analysis demonstrates that reaction sequences 1-C_4H_8 → aC_3H_5 → aC_3H_4 → pC_3H_4 → C_2H_2, 2-C_4H_8 → saxC_4H_7 → 1,3-C_4H_6 → C_2H_3 → C_2H_2 and *i*C_4H_8 → *i*-C_4H_8 → *i*C_4H_7 → aC_3H_4 → pC_3H_4 → C_2H_2 are the major decomposition pathways of 1-butene, 2-butene and *i*-butene, respectively.

1. Introduction

Butanols and large alkanes such as *n*-heptane, isooctane and *n*-decane are attracting more and more concerns for both research purpose and practical use. Since their primary decomposition can produce high concentrations of alkenes, the development of kinetic models of these important fuels requires accurate alkene chemistry. Among alkenes, butene is of special interest considering the important role in the decomposition of butanols and large alkanes and insufficiently known chemistry. On the other hand, butene is the smallest alkene with isomeric structures (including linear 1-butene and 2-butene, and branched *i*-butene) and the largest gaseous alkene at room temperature, which facilities the studies of kinetic issues of isomeric fuel structures.

In the past four decades, some studies on pyrolysis, oxidation, and combustion of 1-butene and *i*-butene have been performed, which are listed in Table 1. For 1-butene, Chakir et al. [1] measured and modeled the oxidation of 1-butene in a jet stirred reactor (JSR). Pitz et al. [2] proposed a mechanism for the oxidation of hydrocarbons including reactions of C4 alkenes. In 2002, Heyberger et al. [3] developed the oxidation model of 1-butene, which reproduced species concentrations in JSR and ignition delay in shock tube very well. For *i*-butene, many studies have been performed on its pyrolysis [4–6], oxidation [6–11], and combustion [12]. The pyrolysis of *i*-butene in the shock tube was investigated by many groups [4–6]. The oxidation of *i*-butene was studied in a turbulent flow reactor [7], shock tube [6,8–10] and JSR [11]. Recently, Dias et al. [12] studied a lean premixed *i*-butene/hydrogen/oxygen/argon flame (ϕ = 0.225) by molecular beam mass spectrometry at low pressure (40 mbar). To the best of our knowledge, no previous study of 2-butene combustion has been reported.

Various conventional diagnostic techniques including ultraviolet (UV) absorption spectroscopy [6], electron-impact ionization mass spectrometry (EIMS) [12] and gas chromatography (GC) [6,11,13] were used in previous studies to detect species in butene combustion. However, the UV absorption spectroscopy can only detect specific species, EIMS is precluded to detect radicals and distinguish isomeric species due to the limited energy resolution of the electron and fragmental interference, and GC cannot detect reactive species such as radicals. Therefore comprehensive detection of combustion or pyrolysis species such as radicals and isomers is desired for the validation of kinetic model.

In this work, we report a pyrolysis study of three butene isomers (1-butene, 2-butene and *i*-butene) in a flow reactor focusing on the decomposition reactions of fuel and recombination

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Table 1
Previous experimental studies on butene isomers.

Year	Temperature range (K)	Pressure	Reaction type	Reactor	References
1-butene					
1989	900–1200	1.5–10 bar	Oxidation	Jet-stirred reactor	[1]
2002	900–1200	1.5–10 bar	Oxidation	Jet-stirred reactor	[3]
2002	1200–1670	6.6–8.9 atm	Oxidation	Shock tube	[3]
2-butene					
–	–	–	–	–	–
i-butene					
1976	1055–1325	292–400 kPa	Pyrolysis	Shock tube	[4]
1986	1080	1 atm	Oxidation	Turbulent flow reactor	[7]
1989	1000–1180	2.5–3 atm	Oxidation	Shock tube	[8]
1992	1100–1900	2.0–4.5 atm	Oxidation	Shock tube	[9]
1998	800–1230	1–10 atm	Oxidation	Jet-stirred reactor	[11]
1998	1230–1930	9.5–10.5 atm	Oxidation	Shock tube	[10]
2003	900–2300	7–400 Torr	Pyrolysis	Shock tube	[5]
2009	1000–1800	1.0–2.7 atm	Pyrolysis, oxidation	Shock tube	[6]
2010	400–1720	40 mbar	Flame	Premixed laminar flame	[12]

reactions to produce cyclic and aromatic species. Comprehensive detection of pyrolysis species, particularly radicals and isomers, was achieved by using synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS) which is benefitting from the advantages of synchrotron VUV light such as broad tunability and high energy resolution. Furthermore, a kinetic model was developed to simulate the pyrolysis of three butene isomers.

2. Experimental method

The experimental work was carried out at National Synchrotron Radiation Laboratory (NSRL) in Hefei, China. The detailed description of synchrotron beamlines used here can be found in our previous work [14,15]. Briefly, two beamlines were used for this study: one utilized undulator radiation covering energy range of 8–17 eV with a gas filter to eliminate high harmonic radiation, while another one used bending magnet radiation covering energy range of 6.5–11.7 eV with LiF window to cut off the high harmonic radiation. Both undulator and bending magnet beamlines used 1 m Seya-Namioka monochromators with resolving powers of 1000 and 500, respectively. But in contrast, the bending magnet beamline can provide higher photon flux at lower energy region.

The pyrolysis apparatus was reported in detail elsewhere [16,17]. In short, the experimental setup consists of three parts: a pyrolysis chamber with a laminar flow reactor electronically heated by a high temperature furnace, a differentially pumped chamber with a molecular-beam sampling system, and a photoionization chamber with a home-made reflectron time-of-flight mass spectrometer with mass resolving power of ~2000. The mixture of each butene isomer and Ar was fed into a 0.6-cm-ID (inner diameter) and 34.0-cm length alumina flow tube with 5.0-cm heated in the high temperature furnace, as shown in Fig. 1. Flow rates of Ar and butenes controlled by mass flow controllers (MKS, USA) were 500 and 20.83 standard cubic centimeters per minute (SCCM), respectively. Thus, the inlet mole fraction of butene isomers was calculated to be 4.0%. In this work, butene isomers were purchased from Nanjing Special Gases Ltd. in China with purities ⩾99.0%. No further purification was carried out. Ar (99.999%) was also purchased from the same vendor.

A quartz cone-like nozzle with a ~500 μm orifice and 40° cone angle at the tip was used to sample the pyrolysis species including the reactants, products, and intermediates. The nozzle orifice was located at 1.0 cm downstream from the outlet of the flow tube, which can avoid secondary reactions after sampling compared with inserting the sampling probe directly into the tube. The sampled pyrolysis species formed a molecular beam in the differentially pumped chamber, and passed into the photoionization chamber through a nickel skimmer. Then, the molecular beam was crossed by the tunable synchrotron VUV light in the ionization region. Photoions produced by the VUV light were drawn out of the photoionization region by a pulse extraction field triggered with a pulse generator (DG 535, SRS, USA), The ion signal was detected by a microchannel plate detector and recorded by a multiscaler (P7888, FAST Comtec, Germany) after a preamplifier (VT120C EG & G, ORTEC, USA) [18].

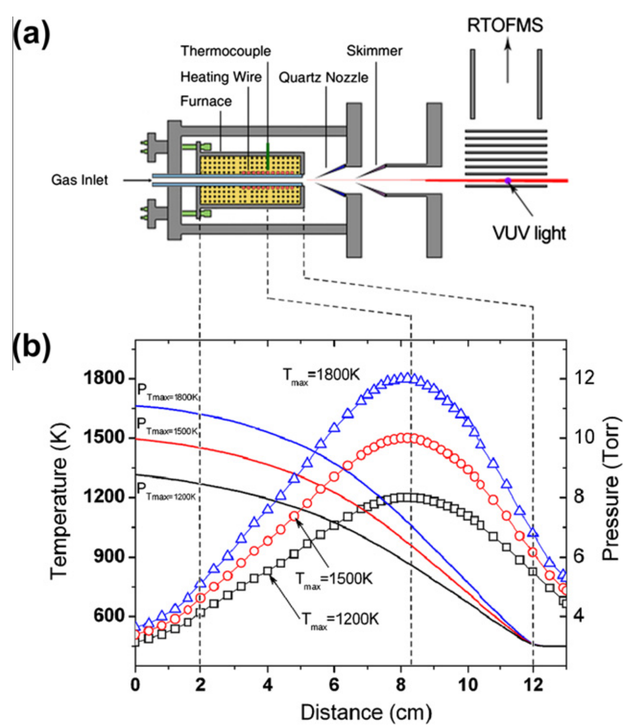

Fig. 1. (a) The schematic diagram of pyrolysis apparatus with molecular-beam sampling mass spectrometer. The red circles inside the furnace denote electronically heating wire with the length of 5.0 cm in this work. A tungsten–rhenium (W–Re) thermocouple that connected to a temperature controller was put close to the middle region of the heating wire. (b) The temperature profiles along the centerline of the flow tube were measured by moving a B-type thermocouple (no shown in the figure) from the tube inlet to the sampling point of the quartz nozzle. Three temperature and pressure profiles are chosen to illustrate variation along the tube. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The experiment was carried out in two modes. (1) The mass spectra can be recorded at a fixed pyrolysis temperature for various photon energies to yield photoionization efficiency (PIE) spectra. In this study, the PIE spectra were measured at the pyrolysis temperature of 1800 K. The ionization energies (IEs) obtained with PIE measurements are used for species identification. (2) The mass spectra can also be recorded at fixed photon energies for various pyrolysis temperatures to yield mole fraction profiles of species versus temperatures. In this study, the mass spectra were measured from 900 to 1900 K at photon energies of 16.64, 12.30, 11.70, 11.00, 10.00, 10.50 and 9.50 eV to obtain near-threshold photoionization. The evaluation method for the mole fraction has been reported in detail previously [16,17]. For species i, the ion signal (S_i) is given by,

$$S_i(T,E) = C \times X_i(T,E) \times \sigma_i(E) \times D_i \times \Phi_p(E) \times \lambda(T), \tag{E1}$$

where C is a proportional constant, $X_i(T,E)$ is the mole fraction of species i at the temperature T and photon energy E, ${}_i(E)$ the photoionization cross section of the species i at the photon energy E, D_i is the mass discrimination factor for the species i, and $\phi_p(E)$ the photon flux. The mole fractions of each species were calculated at various photon energies. For example, the mole fractions of C_3H_6 were calculated at 10.00, 10.50, 11.00 and 11.70 eV, respectively. And it is found that the mole fraction data are consistent with each other at 10.00, 10.50 and 11.00 eV, and the fragment of 1-C_4H_8 exists at 11.70 eV. Generally, the mole fraction profiles calculated at the lowest energy were chosen. If the signal is too weak at the lowest energy, a higher energy is used instead. The total C and H balances for measured species are within 10% uncertainty in comparison to carbon and hydrogen input.

To simulate the pyrolysis experiment, the temperature and pressure distributions inside the flow tube should be considered, which are given below.

2.1. Temperature profiles inside the flow tube

As shown in Fig. 1a, a tungsten–rhenium thermocouple was put close to the middle region of the heating wire to monitor the outside temperature of the flow tube, named $T_{outside}$. Before the experiment, another thermocouple (B-type) was put inside the flow tube to measure the temperature profiles along the flow tube with the Ar flow rate of 500 SCCM. The position of the B-type thermocouple can be controlled by a feedthrough outside the pyrolysis chamber. The temperature uncertainty inside the flow tube is within ±50 K by repetitious measurements and different types of thermocouples. In this work, we chose different $T_{outside}$s and measured the temperature profiles along the flow tube. Hence the relationship between the $T_{outside}$s and the temperature profiles inside the flow tube was obtained. Figure 1b shows the temperature profiles along the flow tube, in which each profile is named by its maximum temperature (T_{max}). In this work, the T_{max}s were used as experimental temperatures. As shown in Fig. 1b, the maximum temperature is located at about 3.7 cm upstream the outlet of the flow tube, which is around the middle of the heating region. These temperature profiles were used in the simulation to predict the mole fractions of pyrolysis species near the orifice of sampling nozzle. The temperature profiles (from T_{max} = 600 to 1900 K) are shown in the Supplemental materials. The temperature profiles between two given temperature profiles can be calculated by interpolation. For example, the temperature profile for T_{max} = 1550 K can be calculated by a relationship below (E2).

$$\text{Profile } T_{max}\ (1550\ \text{K}) = [\text{Profile } T_{max}\ (1500\ \text{K}) + \text{Profile } T_{max}\ (1600\ \text{K})]/2 \tag{E2}$$

2.2. Pressure profiles inside the flow tube

To reduce collisions of the pyrolysis species and detect the primary and secondary products including free radicals, the pressure at the outlet of the flow tube was kept at 3 Torr in this work. The Reynolds number (Re) of the flow was calculated to be less than 154 over the temperature range from T_{max} = 600 to 1900 K. Thus it is a laminar flow in view of Re < 2000. The pressure distribution along the flow tube can be calculated by one dimensional compressible flow equation (E3). The flow conditions are evaluated by using pure Ar, since Ar concentration (96%) is much higher than that of butene isomers (4%).

$$v dv + dp/\rho + dF = 0 \tag{E3}$$

Here v, p, ρ and F are velocity, density, pressure and frictional resistance, respectively.

$$dF = 2f v dl/D \tag{E4}$$

$$f = 16/\mathrm{Re} \tag{E5}$$

$$\mathrm{Re} = v\rho D/\mu \tag{E6}$$

$$\mu = \mu_0 (T/298)^{1.5}(298+s)/(T+s) \tag{E7}$$

Frictional resistance can be calculated by (E4), (E5), (E6), (E7), where f, D, μ and dl are friction factor, internal diameter (0.006 m), viscosity and the distance upstream from the outlet of the flow tube. For Ar, μ_0 and s are 2.125×10^{-5} and 110.56, respectively.

$$v = G/\rho = GRT/pM \tag{E8}$$

Here G and M are the mass flow rate (0.547 kg s^{-1} m^{-2}) and molar mass of Ar (0.040 kg·mol^{-1}), respectively. Take (E4) and (E8) into (E3) to get equation below,

$$(1/p - \rho/G^2)dp = dT/T + 2fdl/D \tag{E9}$$

The pressure at the outlet of the flow tube was kept at 3 Torr and the temperatures along the flow tube were measured. Thus the pressure distribution inside the flow tube can be calculated by integral of (E9). Figure 1b shows the pressure distribution along the flow tube. As shown in Fig. 1b, the pressure decreases more quickly at high temperature region, since higher temperature leads to higher velocity (E8) and frictional resistance (E4). The pressure profiles from T_{max} = 600 to 1900 K were used in the simulation and given in the Supplemental materials.

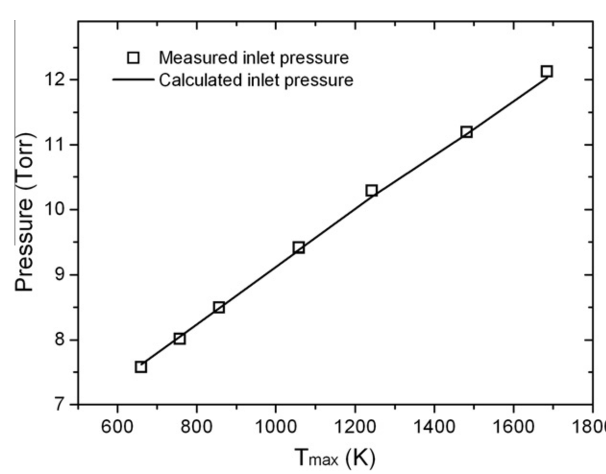

Fig. 2. Measured (open symbols) and calculated (solid lines) pressures at the inlet of the flow tube.

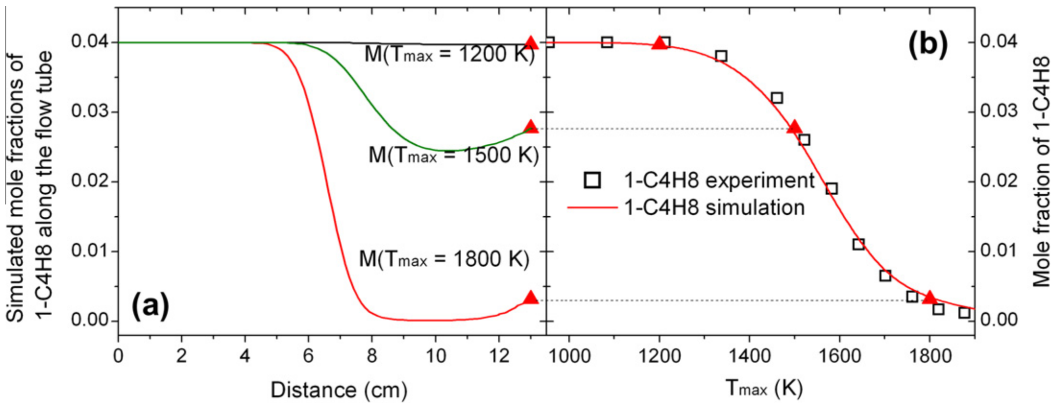

Fig. 3. (a) Simulated mole fractions of 1-C_4H_8 along the flow tube at T_{max} = 1200, 1500 and 1800 K in the pyrolysis of 1-C_4H_8. (b) Experimental and simulated mole fractions of 1-C_4H_8 at the sample position (13 cm from the inlet) over temperature range from T_{max} = 950 to 1900 K.

To validate the calculation of the pressure, the pressures at the inlet of the flow tube (34.0 cm away from the outlet) were measured over the temperature range from T_{max} = 660 to 1685 K. As shown in Fig. 2, the calculated pressures at the inlet of the flow tube are in good agreement with the experimental measurements.

The residence time is another key factor in the flow tube experiment. In this work, it is defined as the time taken by a gas element to travel from 0 to 13 cm. Thus the residence time is calculated to be 1.25–1.33 ms with the plug flow code of the Chemkin program basing on the pressures and temperatures measured above. However, this value depends on the T_{max} and fuel.

3. Kinetic modeling

The simulation work were performed with Plug Flow Code using Chemkin-PRO software [19]. Recognizing the distributions of the pressure and temperature along the flow tube axis in our experiments discussed above, we used the local pressures and temperatures instead of constant values as input parameters in the simulation. This modification is especially significant for pyrolysis and oxidation in the low-pressure flow tube with short heating region, where the axial distributions of pressure and temperature cannot be treated as constant values.

Here we take 1-C_4H_8 as an example to introduce the comparison between the simulation and experiment. Figure 3a shows the simulated mole fractions of 1-C_4H_8 along the flow tube with three T_{max}s in the pyrolysis of 1-C_4H_8. As shown in Fig. 1, the pyrolysis species were sampled at 13 cm. Therefore only the simulation results at 13 cm are used to compare with the experimental results with different T_{max}s (see Fig. 3b). The temperature decreases after 8 cm, shown in Fig. 1. But there is an increase in simulated concentrations of 1-C_4H_8 between 11 and 13 cm for the 1500 K experiment and 12 and 13 cm for the 1800 K experiment, respectively, as shown in Fig. 3a. This is due to that aC_3H_5 and CH_3 radicals recombine to produce 1-C_4H_8 at the low temperature region near the outlet of the flow tube.

The chemical kinetic mechanism was mainly adapted from USC Mech II [20] which has been validated for several C0—C4 fuels [21–25]. Furthermore, rate constants of many reactions were updated with recently calculated or experimental results, especially the reactions related to three butene isomers. The present model consists of 76 species and 232 elementary reactions. Table 2 lists some selected reactions and their rate constants. Most of the thermodynamic and transport data were taken from USC Mech II [20], while other data were taken from the compilation of Burcat and Ruscic [26]. The full mechanism is included in the Supplemental materials.

3.1. Decomposition of 1-butene

1-Alkenes are important intermediate species in the combustion of normal alkanes such as *n*-heptane and *n*-decane. High temperature chemical kinetics of 1-alkenes, including 1-butene [3], 1-pentene [27–29], 1-hexene [30–35], have been investigated experimentally and theoretically. However, most of them focused on the oxidation reactions [3,27–32,34].

Two reactions involving C—C bond breaking of 1-butene (R1) and (R2) were included in this work. The pressure-dependent rate constants of (R1) and (R2) were adapted from Tsang et al.'s work [36,37], which were also adopted by USC Mech II [20].

$$\mathrm{aC_3H_5 + CH_3 = 1\text{-}C_4H_8} \quad (R1)$$

$$\mathrm{C_2H_3 + C_2H_5 = 1\text{-}C_4H_8} \quad (R2)$$

$$\mathrm{1 - C_4H_8 + H = saxC_4H_7 + H_2} \quad (R5)$$

$$\mathrm{1 - C_4H_8 + H = C_4H_7 + H_2} \quad (R6)$$

$$\mathrm{C_4H_7 = 1,3\text{-}C_4H_6 + H} \quad (R11)$$

$$\mathrm{C_4H_7 = C_2H_4 + C_2H_3} \quad (R12)$$

$$\mathrm{C_4H_7 = saxC_4H_7} \quad (R13)$$

In this model, two types of H abstraction reactions of 1-butene (R5) and (R6) producing 1-buten-3-yl radical (saxC_4H_7) and 1-buten-4-yl radical (C_4H_7) were considered. Recently, Miyoshi [38] studied the unimolecular decomposition (R11) and (R12) and isomerization reactions (R13) of C_4H_7 radical using transition state theory based on the quantum chemical calculations at the CBS-QB3 level. These rate constants were also adopted by this model.

The rate parameters of some reactions of 1-C_4H_8 which have not been investigated were assigned from those of analogous reactions.

3.2. Decomposition of 2-butene

To the best of our knowledge, no mechanism has been reported to simulate the 2-butene pyrolysis. Thus we proposed a sub-mechanism of 2-butene to simulate our experimental results. The rate constants of most reactions were assigned to those of analogous reactions.

The 1,3-H shift reaction of 2-butene (R15), which leads to the formation of aC_3H_5 and CH_3, was proposed to explained the high concentration of aC_3H_5. Detailed discussions of this reaction are given in the Section 4.

Table 2
Rate constants for selected reactions in the present pyrolysis model of three butene isomers, $k = AT^n\exp(-E/RT)$. The units are in cm^3, mol, s, cal.

	Selected reactions	A	n	E	References
Reactions of 1-C_4H_8					
1	$aC_3H_5 + CH_3$ (+M) = 1-C_4H_8 (+M)	1.00×10^{14}	−0.32	−262.3	[20]
	Low pressure limit	3.91×10^{60}	−12.81	6250	
	Troe parameters: 0.104 1606 60,000 6118.4				
2	$C_2H_3 + C_2H_5$ (+M) = 1-C_4H_8 (+M)	1.50×10^{13}	0.0	0.0	[20]
	Low pressure limit	1.55×10^{56}	−11.79	8984.5	
	Troe parameters: 0.198 2278 60,000 5723.2				
3	saxC_4H_7 + H = 1-C_4H_8	5.00×10^{13}	0.0	5000	Est.
4	C_4H_7 + H (+M) = 1-C_4H_8 (+M)	3.60×10^{13}	0.0	0.0	[20]
	Low pressure limit	3.01×10^{48}	−9.32	5833.6	
	Troe parameters: 0.498 1314 1314 50,000				
5	1-C_4H_8 + H = saxC_4H_7 + H_2	1.30×10^{6}	2.4	4470	[52]
6	1-C_4H_8 + H = C_4H_7 + H_2	1.30×10^{6}	2.4	4470	[33][a]
7	1-C_4H_8 + H = C_3H_6 + CH_3	3.20×10^{22}	−2.39	11,180	[33]
8	1-C_4H_8 + H = C_2H_4 + C_2H_5	8.80×10^{16}	−1.05	6461	[b]
9	1-C_4H_8 + CH_3 = saxC_4H_7 + CH_4	2.82×10^{0}	3.60	7153	[c]
10	saxC_4H_7 (+M) = C_4H_6 + H (+M)	4.70×10^{8}	1.32	44,698	[20]
	Low pressure limit	4.60×10^{-37}	15.37	−603	
11	C_4H_7 (+M) = C_4H_6 + H (+M)	3.16×10^{13}	0.0	17,160	[38]
	Low pressure limit	1.47×10^{28}	−3.683	13,890	
	Troe parameters: 0.81 50 1150 50,000				
12	C_4H_7 (+M) = C_2H_4 + C_2H_3 (+M)	1.78×10^{14}	0.0	18,590	[38]
	Low pressure limit	4.20×10^{37}	−6.262	18,970	
	Troe parameters: 0.81 50 1670 50,000				
13	C_4H_7 (+M) = saxC_4H_7 (+M)	5.75×10^{12}	0.0	30,400	[38]
	Low pressure limit	1.17×10^{15}	0.0	8170	
	Troe parameters: 0.31 50 100,000 50,000				
Reactions of 2-C_4H_8					
14	$CH_3CHCH + CH_3$ (+M) = 2-C_4H_8 (+M)	5.00×10^{13}	0.0	0.0	[d]
	Low pressure limit	8.54×10^{58}	−11.94	9770	
	Troe parameters: 0.175 1340.6 60,000 10139.8				
15	2-C_4H_8 = aC_3H_5 + CH_3	7.50×10^{66}	−15.6	97,300	See text
16	2-C_4H_8 = saxC_4H_7 + H	1.06×10^{85}	−20.03	132,787	[e]
17	2-C_4H_8 + H = saxC_4H_7 + H_2	3.16×10^{6}	2.5	6756	[f]
18	2-C_4H_8 + H = C_3H_6 + CH_3	3.46×10^{17}	−1.05	6461	[g]
19	2-C_4H_8 + H = CH_3CCHCH_3 + H_2	1.32×10^{6}	2.53	12,240	[h]
20	2-C_4H_8 + CH_3 = saxC_4H_7 + CH_4	4.40×10^{0}	3.50	5675	[i]
Reactions of iC_4H_8					
21	iC_4H_8 (+M) = CH_3CCH_2 + CH_3 (+M)	2.50×10^{13}	0.0	0.0	[20]
	Low pressure limit	4.27×10^{58}	−11.94	9770	
	Troe parameters: 0.175 1341 60,000 10,140				
22	iC_4H_7 + H (+M) = iC_4H_8 (+M)	2.00×10^{14}	0.0	0.0	[20]
	Low pressure limit	1.33×10^{60}	−12.0	5968	
	Troe parameters: 0.02 1097 1097 6859				
23	iC_4H_8 + H = iC_4H_7 + H_2	1.72×10^{14}	0.0	8000	[6]
24	iC_4H_8 + H = C_3H_6 + CH_3	8.80×10^{16}	−1.05	6461	[j]
25	iC_4H_8 + H = iC_4H_7-1 + H_2	1.20×10^{14}	0.0	13,000	[6]
26	iC_4H_8 + CH_3 = iC_4H_7 + CH_4	4.40×10^{0}	3.50	5675	[k]
27	iC_4H_7-1 = pC_3H_4 + CH_3	1.20×10^{13}	0.0	37,000	[6][l]
28	iC_4H_7 = aC_3H_4 + CH_3	5.00×10^{11}	0.0	51,000	[6][n]
29	iC_4H_7 = saxC_4H_7	6.00×10^{13}	0.0	70,000	See text
Reactions of 1,3-C_4H_6					
30	1,3-C_4H_6 = H_2CC + C_2H_4	1.00×10^{13}	0.0	85,000	[m]
31	1,3-C_4H_6 + H = C_2H_4 + C_2H_3	1.46×10^{30}	−4.34	21,647	[20]
32	1,3-C_4H_6 + H = nC_4H_5 + H_2	1.33×10^{6}	2.53	12,240	[20]
33	1,3-C_4H_6 + H = iC_4H_5 + H_2	6.65×10^{5}	2.53	9240	[20]
Reactions of C_3H_6					
34	aC_3H_5 + H (+M) = C_3H_6 (+M)	2.00×10^{14}	0.0	0.0	[20]
	Low pressure limit	1.33×10^{60}	−12.0	5968	
	Troe parameters: 0.020 1096.6 1096.6 6859.5				
35	C_3H_6 + H = aC_3H_5 + H_2	1.73×10^{5}	2.50	2490	[20]
36	C_3H_6 + H = CH_3CCH_2 + H_2	4.00×10^{5}	2.5	9790	[20]
37	C_3H_6 + H = CH_3CHCH + H_2	8.04×10^{5}	2.5	12,283	[20]
38	C_3H_6 + H = C_2H_4 + CH_3	8.80×10^{16}	−1.05	6461	[20]
Reactions of C_3H_5					
39	aC_3H_4 + H = aC_3H_5	9.60×10^{60}	−14.67	26,000	[20][n]
40	pC_3H_4 + H = aC_3H_5	1.10×10^{59}	−14.56	28,100	[20][n]
41	aC_3H_5 + H = aC_3H_4 + H_2	9.56×10^{3}	2.80	3291	[53]
42	aC_3H_5 + CH_3 = aC_3H_4 + CH_4	3.00×10^{12}	−0.32	−131	[33]
43	aC_3H_5 + aC_3H_5 = pC_3H_4 + C_3H_6	8.43×10^{10}	0.0	−263	[33]
44	CH_3CHCH + H = aC_3H_5 + H	1.47×10^{13}	0.26	4103	[o]
45	CH_3CCH_2 + H = aC_3H_5 + H	1.47×10^{13}	0.26	4103	[o]

(*continued on next page*)

Table 2 (*continued*)

	Selected reactions	A	n	E	References
46	$H + pC_3H_4 = CH_3CHCH$	3.38×10^{50}	−12.75	7722	[42]
47	$CH_3 + C_2H_2 = CH_3CHCH$	1.70×10^{49}	−12.3	16,642	[42]
48	$H + aC_3H_4 = CH_3CCH_2$	4.00×10^{52}	−13.1	14,472	[42]
49	$H + pC_3H_4 = CH_3CCH_2$	8.00×10^{52}	−12.69	14,226	[42]
Reactions of C_3H_4					
50	$pC_3H_4 = aC_3H_4$	6.40×10^{61}	−14.59	88,200	[44]
51	$H + aC_3H_4 = H + pC_3H_4$	1.47×10^{13}	0.26	4103	[42]
52	$aC_3H_4 = C_3H_3 + H$	5.02×10^{50}	−10.96	102,317	[33][p]
53	$pC_3H_4 = C_3H_3 + H$	2.12×10^{44}	−9.06	100,390	[33][p]
54	$aC_3H_4 + H = C_3H_3 + H_2$	6.60×10^{03}	3.1	5522	[42]
55	$pC_3H_4 + H = C_3H_3 + H_2$	3.57×10^{04}	2.8	4821	[42]
56	$H + aC_3H_4 = CH_3 + C_2H_2$	2.72×10^{09}	1.2	6834	[42]
57	$H + pC_3H_4 = C_2H_2 + CH_3$	3.89×10^{10}	0.989	4114	[42]
Reactions of C_5H_6 and C_6H_6					
58	$C_5H_5 + H\ (+M) = C_5H_6\ (+M)$	1.00×10^{14}	0.0	0.0	[20]
	Low pressure limit	4.40×10^{80}	−18.28	12,994	
	Troe parameters: 0.175 1340.6 60,000 10139.8				
59	$C_5H_5\ (+M) = C_2H_2 + C_3H_3\ (+M)$	6.31×10^{13}	−0.075	62,300	[54]
	Low pressure limit	1.00×10^{45}	8.4	47,500	
60	$aC_3H_5 + C_2H_2 = lC_5H_7$	8.38×10^{30}	−6.2	12,824	[20]
61	$iC_4H_5 + CH_3 = lC_5H_7 + H$	1.00×10^{13}	0.0	0.0	Est.
62	$C_5H_6 + H = lC_5H_7$	8.27×10^{126}	−32.3	82,348	[20]
63	$aC_3H_5 + C_2H_3 = lC_5H_7 + H$	1.00×10^{13}	0.0	0.0	[20]
64	$C_5H_6 + H = C_5H_5 + H_2$	3.00×10^{14}	0.0	6640	[51]
65	$C_5H_6 + H = C_2H_2 + aC_3H_5$	7.74×10^{36}	−6.2	32,890	[20]
66	$nC_4H_3 + C_2H_3 = C_6H_6$	2.87×10^{41}	0.0	817	[55]
67	$iC_4H_5 + C_2H_2 = C_6H_6 + H$	3.00×10^{11}	0.0	14,900	[56]
68	$aC_3H_4 + C_3H_3 = C_6H_6 + H$	2.20×10^{11}	0.0	2000	[56]
69	$C_4H_4 + C_2H_2 = C_6H_6$	4.47×10^{11}	0.0	30,010	[50]
70	$nC_4H_5 + C_2H_2 = C_6H_6 + H$	2.10×10^{15}	−1.1	4800	[20]
71	$nC_4H_5 + C_2H_3 = C_6H_6 + H_2$	1.84×10^{-13}	7.1	−3611	[20]
72	$C_3H_3 + C_3H_3 = C_6H_6$	3.50×10^{66}	−15.9	27,529	[43][p]
73	$C_3H_3 + C_3H_3 = C_6H_6$	2.40×10^{35}	−7.4	5058	[43][p]

[a] Refer to the reaction $C_6H_{12} + H = C_6H_{11}\text{-}5 + H_2$ [33].
[b] Refer to the reaction $C_3H_6 + H = C_2H_4 + CH_3$[20].
[c] Refer to the reaction $C_6H_{12} + CH_3 = saxC_6H_{11} + CH_4$ [33].
[d] Refer to the reaction $C_2H_3 + CH_3\ (+M) = C_3H_6\ (+M)$[20].
[e] Refer to the reaction $C_3H_6 = aC_3H_5 + H$ [33], increase by a factor of two considering two methyl group in 2-butene and reduced by a factor of 5 considering the pressure fall-off.
[f] Refer to the reaction $1\text{-}C_6H_{12} + H = saxC_6H_{11} + H_2$ [33].
[g] Refer to the reaction $C_3H_6 + H = C_2H_4 + CH_3$ [20], increase by a factor of 2 considering two methyl group in 2-butene.
[h] Refer to the reaction $C_2H_4 + H = C_2H_3 + H_2$ [20].
[i] Refer to the reaction $C_3H_6 + CH_3 = aC_3H_5 + CH_3$ [20], increase by a factor of 2 considering two methyl group in 2-butene.
[j] Refer to the reaction $C_3H_6 + H = C_2H_4 + CH_3$ [20], increase by a factor of 2 considering two methyl group in iC_4H_8.
[k] Refer to the reaction $C_3H_6 + CH_3 = aC_3H_5 + CH_4$ [20], increase by a factor of 2 considering two methyl group in iC_4H_8.
[l] Reduced by a factor of 20 considering the pressure fall-off.
[m] The threshold energy 85 kcal/mol is from the calculation of Lee et al.[57] with estimated pre-exponential factor.
[n] Reduced by a factor of 10 considering the pressure fall-off.
[o] Refer to the reaction $aC_3H_4 + H = pC_3H_4 + H$ [20].
[p] Reduced by a factor of 5 considering the pressure fall-off.

$$2\text{-}C_4H_8 = CH_3CHCH + CH_3 \quad (R14)$$

$$2\text{-}C_4H_8 = aC_3H_5 + CH_3 \quad (R15)$$

$$C_3H_6 + H = aC_3H_5 + H_2 \quad (R35)$$

$$CH_3CHCH + H = aC_3H_5 + H \quad (R44)$$

3.3. Decomposition of i-butene

Several groups have studied the pyrolysis [5,6], oxidation [6,9,11] and flame [12] mechanism of *i*-butene. In 1992, Curran et al. [9] proposed a sub-mechanism of *i*-butene to simulate the ignition delay time of *i*-butene. In their sub-mechanism, the H abstractions from the methyl group by H and CH_3 (R23) and (R26) were considered. And the produced iC_4H_7 further dissociates via β-scission reaction (R28). In Dagaut's model [11], the unimolecular reactions of iC_4H_8 (R21) and (R22) and the C—C fission by H attack reactions (R24) were included in this work.

$$iC_4H_8 = CH_3CCH_2 + CH_3 \quad (R21)$$

$$iC_4H_8 = iC_4H_7 + H \quad (R22)$$

$$iC_4H_8 + H = iC_4H_7 + H_2 \quad (R23)$$

$$iC_4H_8 + H = C_3H_6 + CH_3 \quad (R24)$$

$$iC_4H_8 + CH_3 = iC_4H_7 + CH_4 \quad (R26)$$

$$iC_4H_7 = aC_3H_4 + CH_3 \quad (R28)$$

Recently, Yasunaga et al. [6] studied the pyrolysis and oxidation of *i*-butene in the shock tube over the temperature range of 1000–1800 K. They developed the model by considering the H

Table 3
List of species detected in the pyrolysis of three butene isomers.

m/z	Formula	Species	IE (eV)			
			Literature[a]	1-Butene[b]	2-Butene	i-Butene
2	H_2	Hydrogen	15.43	√[c]	√	√
15	CH_3	Methyl radical	9.84	9.79	9.82	9.79
16	CH_4	Methane	12.61	√	√	√
26	C_2H_2	Acetylene	11.40	11.37	11.37	11.34
27	C_2H_3	Vinyl radical	8.25	–[d]	–	–
28	C_2H_4	Ethylene	10.51	10.49	10.49	10.49
29	C_2H_5	Ethyl radical	8.12	–	–	–
30	C_2H_6	Ethane	11.52	√	√	√
39	C_3H_3	Propargyl radical	8.67	8.69	8.72	8.72
40	C_3H_4	Propyne	10.36	10.35	10.35	10.35
	C_3H_4	Allene	9.69	9.74	9.72	9.79
41	C_3H_5	Allyl radical	8.18	8.15	8.17	8.15
42	C_3H_6	Propylene	9.73	9.70	9.73	9.76
50	C_4H_2	Diacetylene	10.17	10.14	10.14	10.14
52	C_4H_4	Vinylacetylene	9.58	9.58	9.58	9.55
54	1,3-C_4H_6	1,3-Butadiene	9.07	9.08	9.08	9.08
56	C_4H_8	2-Butene	9.11	No	9.13	No
	C_4H_8	i-Butene	9.22	No	No	9.21
	C_4H_8	1-Butene	9.55	9.55	No	No
66	C_5H_6	1,3-Cyclopentadiene	8.57	8.57	8.59	8.60
78	C_6H_6	Benzene	9.24	9.21	9.24	9.24

[a] Reference to NIST Chemistry Webbook.
[b] The uncertainty for IE is ±0.05 eV in this work.
[c] Decided by MS.
[d] Trace level.

abstraction from other carbon atom of *i*-butene, which produces iC_4H_7-1 (R25).

$$iC_4H_8 + H = iC_4H_7\text{-}1 + H_2 \tag{R25}$$

High concentration of 1,3-C_4H_6 was observed unexpectedly in the *i*-butene pyrolysis in this work. In Yasunaga's iC_4H_8 model [6], they supposed iC_4H_7 would isomerize to a liner C_4H_7 radical ($CH_2{=}C*CH_2CH_3$). Similarly, the isomerization reaction from iC_4H_7 to saxC_4H_7 (R29) was proposed in the present model, which could explain the high concentration of 1,3-C_4H_6, though detailed calculations are still needed.

$$iC_4H_7 = \text{sax}C_4H_7 \tag{R29}$$

3.4. Reactions of C4 and C3 species

Most reactions of C4 species were taken from USC Mech II [20], which have been validated by the pyrolysis and oxidation experiments of 1,3-C_4H_6 [23]. The sub-mechanism of C_4H_7-2 ($CH_2{=}C*{-}CH_2{-}CH_3$) and $CH_3{-}C*{=}CH{-}CH_3$ radicals in the 1,3-C_4H_6 model of Gueniche et al. [39] was also included in this model.

The reactions of C3 species have been investigated by numerous experimental and theoretical studies [25,40–46]. In 1999, Davis et al. [44] investigated the reactions relevant to propyne (pC_3H_4) pyrolysis using ab initio calculations and RRKM theory. The pressure-dependent rate constants of these reactions were obtained over the temperature range 300–2500 K and the pressure range 0.1–100 atm. Recently, Miller et al. [42] calculated the dissociation pathways of aC_3H_5, CH_3CHCH and CH_3CCH_2. These rate constants are used in the present model.

3.5. Reactions of 1,3-cyclopentadiene and benzene

1,3-Cyclopentadiene (C_5H_6) and benzene (C_6H_6) were detected in the pyrolysis of three C_4H_8 isomers. To simulate their concentration profiles, the reactions of C_5H_6 and C_6H_6 were included in the present model. The reactions were taken from the pyrolysis mechanism of toluene [47], which has also been validated by flame data of toluene [48] and ethylbenzene [49]. The sub-mechanism of C_5H_6 in the toluene pyrolysis mechanism [47] was taken from the USC Mech II [20], the low-pressure soot model developed by Richter et al. [50] and the C_5H_6 model proposed by Bacskay and Mackie [51]. C2 + C4 and C3 + C3 reactions (R65)–(R72) have been included in the present model to predict the formation of C_6H_6.

4. Results and discussion

As listed in Table 3, about 20 species with m/z = 2–78 were observed in the pyrolysis of each butene isomer. For the trace species, their IEs are not listed in the table. Several radicals and isomers were identified. For example, methyl and propargyl radicals were detected, while C_3H_4 (m/z = 40) are distinguished as propyne and allene isomers.

The experimental mole fraction profiles (open symbols) and modeling results (solid symbols plus dash lines) of pyrolysis species of 1-butene, 2-butene and *i*-butene are shown in Figs. 4–6, respectively. In general, the modeling results are in good agreement with the experimental measurements. Reaction flux analysis was performed at 1800 K to discuss the decomposition pathways of 1-butene, 2-butene and *i*-butene, as shown in Fig. 7a–c, respectively. The thickness of the arrow in Fig. 7 reflects the flux of corresponding reaction(s). Figures 8 and 9 show sensitivity analysis of aC_3H_5 in the pyrolysis of 2-butene and 1,3-C_4H_6 in the pyrolysis of *i*-butene at 1800 K, respectively.

4.1. 1-Butene pyrolysis

Figure 4 shows the experimental (open symbols) and modeling (solid symbols plus dash lines) mole fraction profiles of pyrolysis species of 1-butene. The decomposition pathways of 1-butene are shown in Fig. 7a based on the reaction flux analysis at 1800 K. The reaction flux analysis indicates that 1-butene mainly decomposes via the β-C—C scission ((R1), 68% consumption) to aC_3H_5 and CH_3 radicals. The H loss on the β-C—H bond (R3) and (R5) also exhibits important influence on the consumption of

1-butene. As shown in Fig. 4, the experimental and predicted mole fractions of 1-butene decrease with the same tendency.

$$saxC_4H_7 + H = 1\text{-}C_4H_8 \quad (R3)$$

$$1\text{-}C_4H_8 + H = saxC_4H_7 + H_2 \quad (R5)$$

$saxC_4H_7$ is another product of the primary step of 1-butene pyrolysis. The further destruction of $saxC_4H_7$ radical is dominated by unimolecular decomposition reactions to form 1,3-C_4H_6 (R10). Most 1,3-C_4H_6 decomposes to C_2H_4 ((R31), 46% consumption) and two C_4H_5 isomers ((R32) and (R33), 26% consumption) by H attack. C_4H_4 and C_4H_2 are produced through the reaction sequence 1,3-$C_4H_6 \rightarrow C_4H_5 \rightarrow C_4H_4 \rightarrow C_4H_3 \rightarrow C_4H_2$. But the reaction flux of this reaction sequence $C_4H_4 \rightarrow C_4H_3 \rightarrow C_4H_2$ is very small, which makes it absent from Fig. 7a. The modeling results of these C4 species are also in consistent with the experimental mole fraction profiles.

$$saxC_4H_7\ (+M) = 1,3\text{-}C_4H_6 + H\ (+M) \quad (R10)$$

$$1,3\text{-}C_4H_6 + H = C_2H_4 + C_2H_3 \quad (R31)$$

$$1,3\text{-}C_4H_6 + H = n\text{-}C_4H_5 + H_2 \quad (R32)$$

$$1,3\text{-}C_4H_6 + H = i\text{-}C_4H_5 + H_2 \quad (R33)$$

Most C_3H_6 is produced by the combination of aC_3H_5 and H ((R34), 51%) and the H attack on 1-C_4H_8 ((R7), 41%). The formation of the intermediate aC_3H_4 is mainly contributed by the unimolecular decomposition of aC_3H_5 radical (R39), while the H abstraction reactions of aC_3H_5 radical by the H and CH_3 radicals (R41) and (R42) also contribute about 35% production.

$$1\text{-}C_4H_8 + H = C_3H_6 + CH_3 \quad (R7)$$

$$aC_3H_5 + H\ (+M) = C_3H_6\ (+M) \quad (R34)$$

$$aC_3H_4 + H = aC_3H_5 \quad (R39)$$

$$aC_3H_5 + H = aC_3H_4 + H_2 \quad (R41)$$

$$aC_3H_5 + CH_3 = aC_3H_4 + CH_4 \quad (R42)$$

About 66% of aC_3H_4 is isomerized to pC_3H_4 via unimolecular reaction (R50) and H attack (R51). These isomerization reactions contribute about 80% on the formation of pC_3H_4. Both aC_3H_4 and pC_3H_4 can be consumed by H attack reaction to produce C_3H_3. But the main decomposed pathway of pC_3H_4 is the reaction with

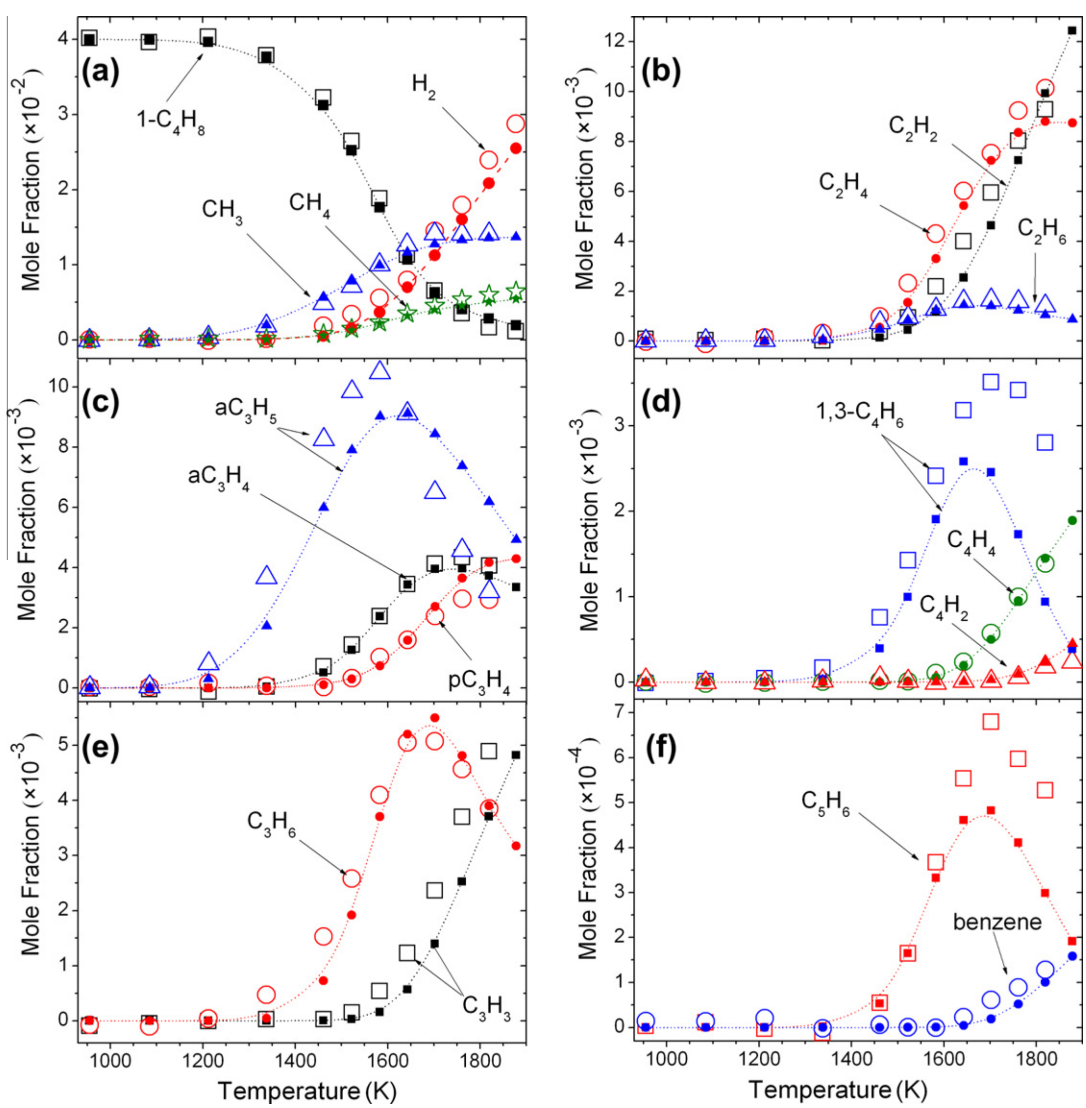

Fig. 4. Experimental mole fraction profiles (open symbols) and modeling results (solid symbols plus dash lines) of pyrolysis species of 1-butene.

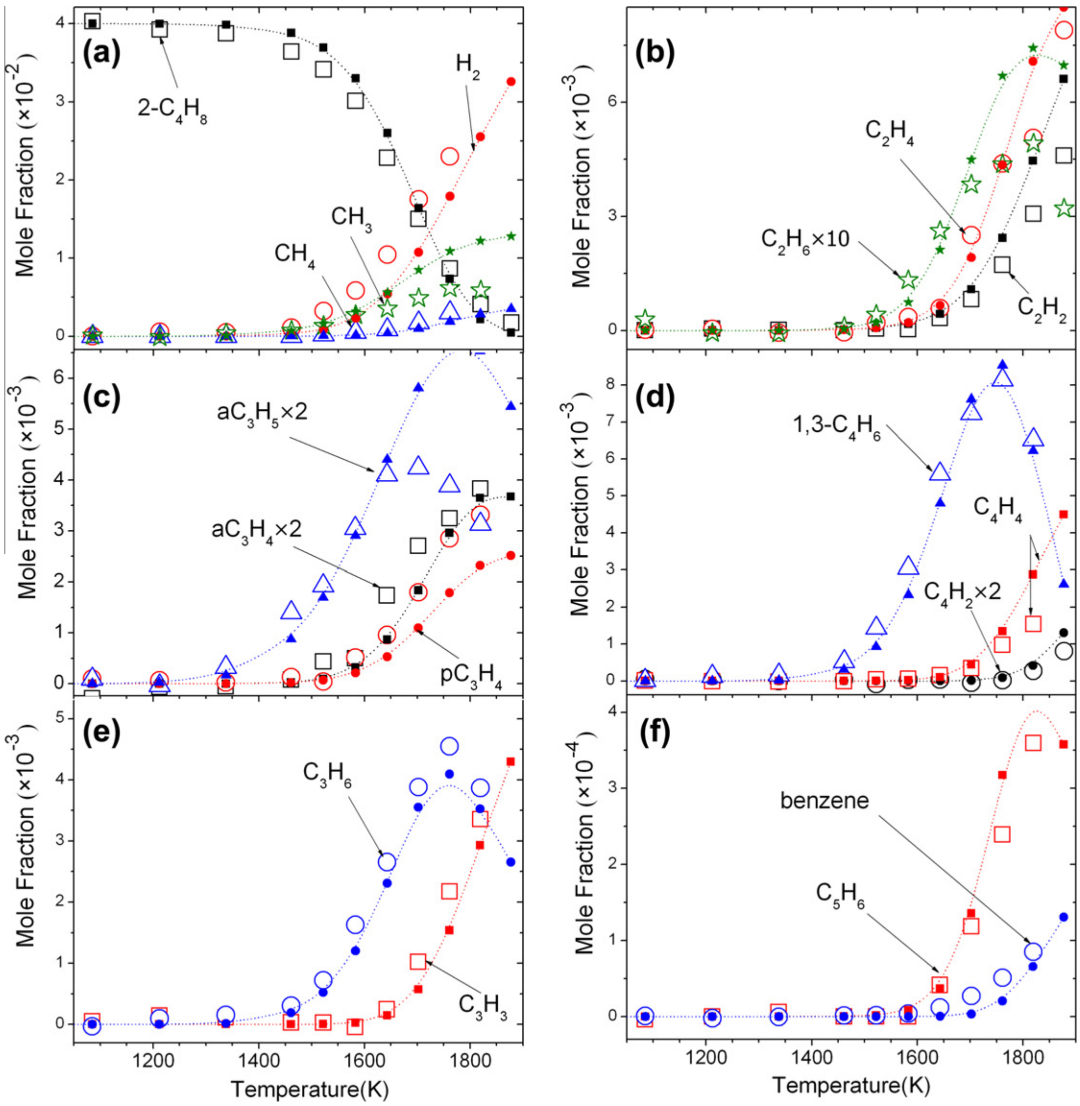

Fig. 5. Experimental mole fraction profiles (open symbols) and modeling results (solid symbols plus dash lines) of pyrolysis species of 2-butene.

H producing C_2H_2 ((R57), 59%). It can be seen in Fig. 4 that the calculated mole fractions of C_3H_6, aC_3H_4, pC_3H_4 and C_3H_3 are in good agreement with the experimental data.

$$pC_3H_4 = aC_3H_4 \quad \text{(R50)}$$

$$H + aC_3H_4 = H + pC_3H_4 \quad \text{(R51)}$$

$$H + pC_3H_4 = C_2H_2 + CH_3 \quad \text{(R57)}$$

Three stable C2 species including C_2H_2, C_2H_4 and C_2H_6 were detected experimentally. The reaction flux analysis shows that most C_2H_6 is produced by the self-combination of CH_3 radical. C_2H_4 and C_2H_2 are mainly produced from C_2H_5 and C_2H_3, respectively. So the reaction C_4H_7 (+M) = C_2H_3 + C_2H_5 (+M) (R12) shows great sensitivity to the formation of C_2H_2 and C_2H_4.

It is concluded that 1-butene is consumed mainly through two reaction sequences 1-C_4H_8 → aC_3H_5 → aC_3H_4 → pC_3H_4 → C_2H_2 and 1-C_4H_8 → $saxC_4H_7$ → 1,3-C_4H_6 → C_2H_3 → C_2H_2, among which the former plays a more significant role.

4.2. 2-Butene pyrolysis

Figure 5 shows the experimental (open symbols) and modeling (solid symbols plus dash lines) mole fraction profiles of pyrolysis species of 2-butene. The decomposition pathways of 2-butene are shown in Fig. 7b. In this model, we considered the C—C scission reaction producing CH_3CHCH and CH_3 radicals (R14), the 1,3-H shift reaction producing aC_3H_5 and CH_3 (R15), the β-C—H scission reaction producing $saxC_4H_7$ (R16), and the reactions between H and 2-C_4H_8 (R17)–(R19) producing $saxC_4H_7$, C_3H_6 and CH_3CCHCH_3.

$$CH_3CHCH + CH_3 = 2\text{-}C_4H_8 \quad \text{(R14)}$$

$$2\text{-}C_4H_8 = aC_3H_5 + CH_3 \quad \text{(R15)}$$

$$2\text{-}C_4H_8 = saxC_4H_7 + H \quad \text{(R16)}$$

$$2\text{-}C_4H_8 + H = saxC_4H_7 + H_2 \quad \text{(R17)}$$

$$2\text{-}C_4H_8 + H = C_3H_6 + CH_3 \quad \text{(R18)}$$

$$2\text{-}C_4H_8 + H = CH_3CCHCH_3 + H_2 \quad \text{(R19)}$$

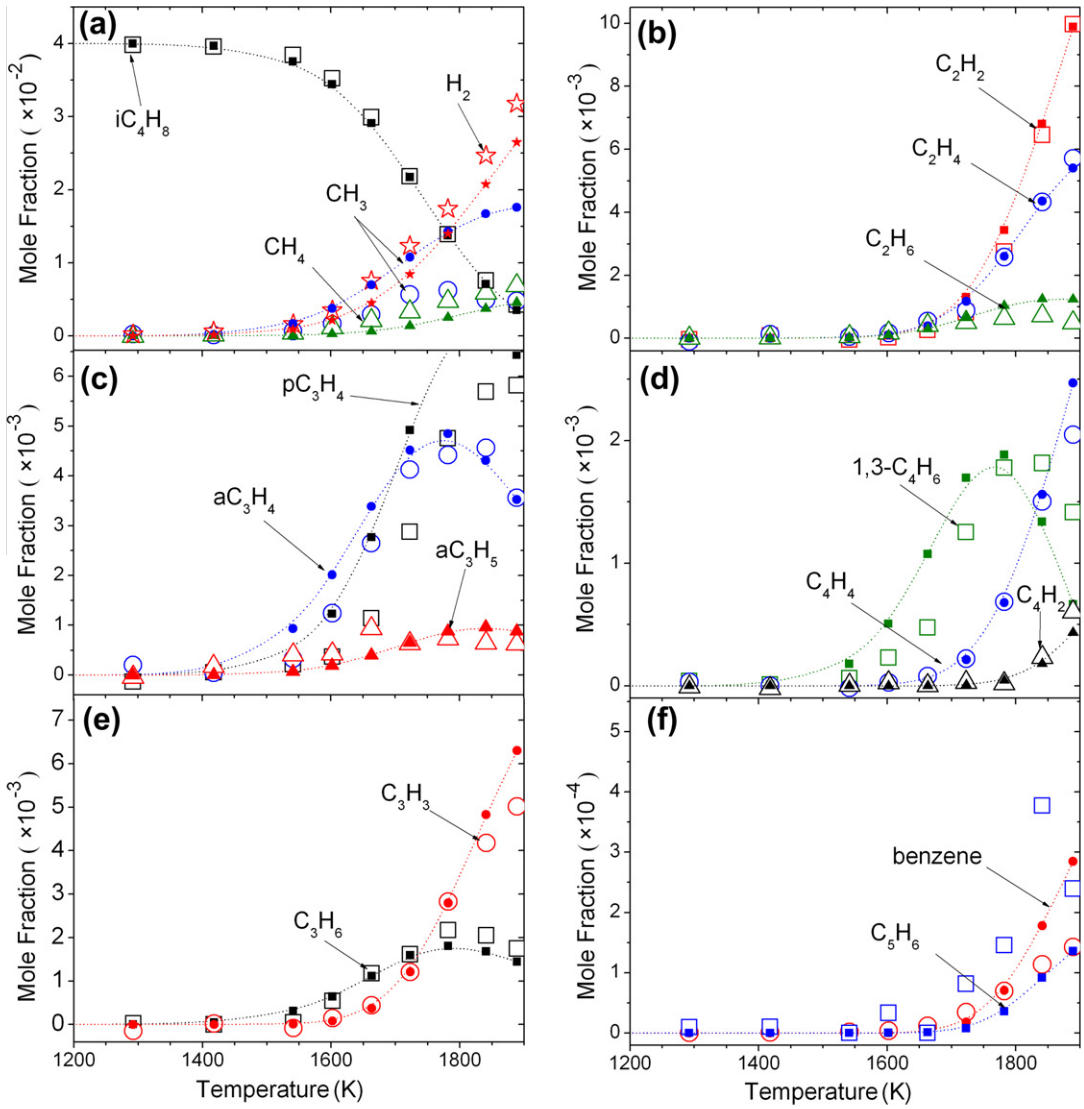

Fig. 6. Experimental mole fraction profiles (open symbols) and modeling results (solid symbols plus dash lines) of pyrolysis species of *i*-butene.

The reaction flux analysis shows that the β-C—H scission (R16) and (R17) is the dominant decomposition pathway of 2-butene (51%). This is different from the dominant contribution of β-C—C scission in the decomposition of 1-butene, which is in accordance with the fact that no β-C—C bond exists in 2-butene. A great amount of $saxC_4H_7$ is produced by the β-C—H scission, leading to the high concentration of 1,3-C_4H_6 which is around three times of that in 1-butene pyrolysis. The main reaction pathways of 1,3-C_4H_6 and smaller C4 species in the pyrolysis of 2-butene are very similar to those in the pyrolysis of 1-butene.

About 13% 2-butene is decomposed via 1,3-H shift reaction to produce aC_3H_5 and CH_3 (R15). This reaction is proposed by our model basing on the high concentration of C_3H_5 species observed in the experiment of 2-butene pyrolysis. CH_3CHCH radical produced by (R14) is decomposed to C_2H_2 and CH_3 rapidly. Even though the H abstraction reaction of C_3H_6 (R35) and isomerization reaction of CH_3CHCH (R44) would contribute to the formation of resonance-stabilized radical aC_3H_5, concentration of aC_3H_5 predicted by the model is still 10 times lower than the experiment results without considering (R15). As shown in Fig. 5, the model can satisfactorily predict the experimental mole fraction profile of aC_3H_5 after adding (R15). And it can be seen from Fig. 8 that (R15) is very important to the formation of aC_3H_5 in 2-C_4H_8 pyrolysis. Meanwhile, reaction $aC_3H_5 + CH_3 = 1\text{-}C_4H_8$ (R1) shows great negative sensitivity coefficients for aC_3H_5. The simulation at $T_{max} = 1800$ K shows that aC_3H_5 and CH_3 combine to produce 1-C_4H_8 at the last 2 cm of the flow tube where the temperature is below 1500 K. Although (R1) has been validated in the 1-C_4H_8 pyrolysis, the large negative sensitivity coefficients of (R1) increase the uncertainty of reaction rate of (R15). Thus the detailed calculation of (R15) is desired.

$$CH_3CHCH + H = aC_3H_5 + H \qquad (R44)$$

$$CH_3 + C_2H_2 = CH_3CHCH \qquad (R47)$$

The reaction pathways of C2 and C3 species in 2-butene pyrolysis are similar to those in 1-butene, which will not be discussed here. In a word, 2-butene is consumed mainly through the reaction sequence 2-C_4H_8 → $saxC_4H_7$ → 1,3-C_4H_6 → C_2H_3 → C_2H_2.

4.3. *i*-Butene pyrolysis

Figure 6 shows the experimental (open symbols) and modeling (solid symbols plus dash lines) mole fraction profiles of pyrolysis species of *i*-butene. The decomposition pathways of *i*-butene are shown in Fig. 7c. The pyrolysis reactions of *i*-butene include the

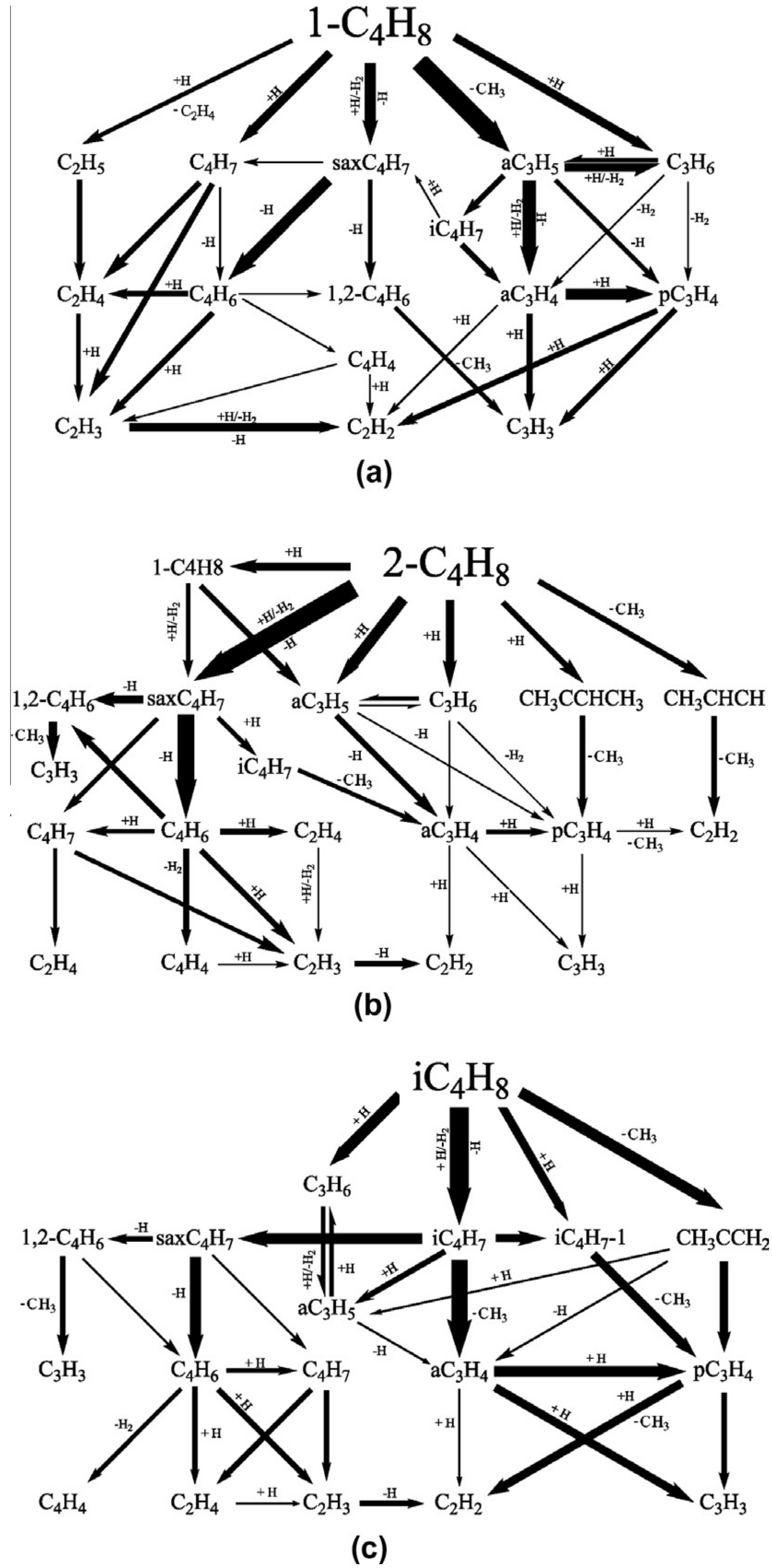

Fig. 7. Decomposition pathways of (a) 1-butene, (b) 2-butene, and (c) *i*-butene at 1800 K. The thickness of the arrows reflects the flux of corresponding reaction(s).

C—C scission (R21) and (R24), β-C—H scission (R22) and (R23), and C—H scission on the double bond (R25). The reaction flux analysis shows that iC_4H_8 is mainly consumed by the β-C—H scission reactions (R22) and (R23) to form iC_4H_7 and is partly converted to CH_3CCH_2 via the C—C scission. The iC_4H_7-1 radical from (R25) is consumed by β-C—C scission to produce pC_3H_4 (R27). But the reaction sequence $iC_4H_8 \rightarrow iC_4H_7\text{-}1 \rightarrow pC_3H_4$ only contributes about 5% to the decomposition of iC_4H_8 at 1800 K.

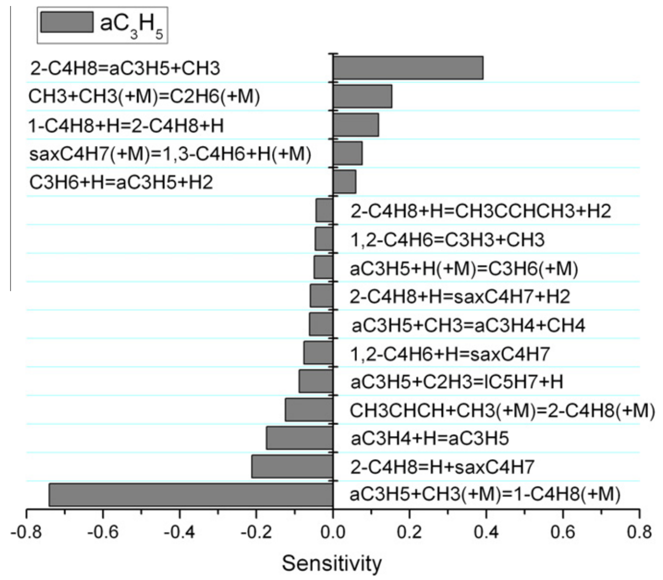

Fig. 8. Sensitivity analysis of aC_3H_5 in the pyrolysis of 2-butene at 1800 K. Only reactions with absolute sensitivity coefficients greater than 0.02 are considered.

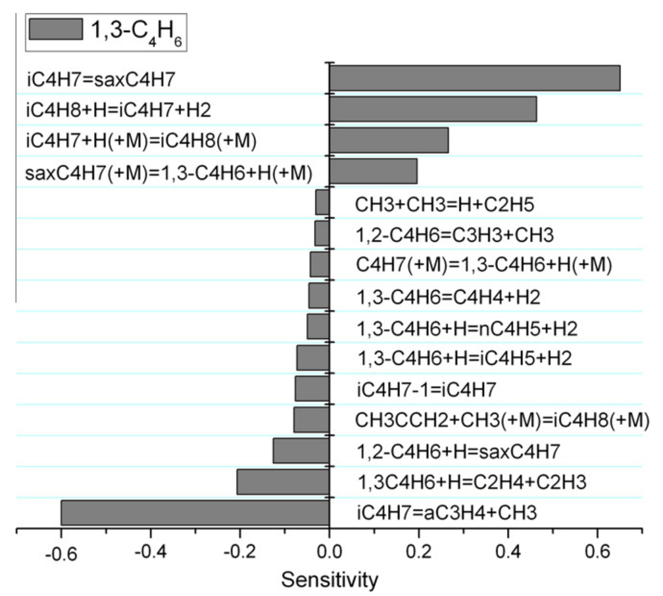

Fig. 9. Sensitivity analysis of $1,3\text{-}C_4H_6$ in the pyrolysis of *i*-butene, at 1800 K. Only reactions with absolute sensitivity coefficients greater than 0.02 are considered.

$$iC_4H_8 = CH_3CCH_2 + CH_3 \tag{R21}$$

$$iC_4H_8 = iC_4H_7 + H \tag{R22}$$

$$iC_4H_8 + H = iC_4H_7 + H_2 \tag{R23}$$

$$iC_4H_8 + H = C_3H_6 + CH_3 \tag{R24}$$

$$iC_4H_8 + H = iC_4H_7\text{-}1 + H_2 \tag{R25}$$

$$iC_4H_7\text{-}1 = pC_3H_4 + CH_3 \tag{R27}$$

iC_4H_7 is mainly decomposed via the β-C—C scission reaction (R28) to produce aC_3H_4 and CH_3, which contributes 90% to the formation of aC_3H_4. As shown in Fig. 6c, the predicted mole fraction profile of aC_3H_4 is in good agreement with the experimental results.

$$iC_4H_7 = aC_3H_4 + CH_3 \tag{R28}$$

Unexpected high concentration of $1,3\text{-}C_4H_6$ (reach to 2×10^{-3}) is observed in the pyrolysis of *i*-butene, as shown in Fig. 6d. The combination reactions of two smaller intermediates in this model, for example, $C_2H_3 + C_2H_3$ or $C_2H_3 + C_2H_4$, cannot explain the experimental observation. In the *i*-butene flame studied by Dias and Vandooren [12], the predicted mole fraction of $1,3\text{-}C_4H_6$ is also about 10 times lower than the experimental results. So some important reactions for $1,3\text{-}C_4H_6$ formation were probably not included in previous iC_4H_8 models. In Yasunaga's iC_4H_8 model [6], they supposed an isomerization reaction from iC_4H_7 to a liner C_4H_7 radical ($CH_2{=}C * CH_2CH_3$). In this experiment, C_4H_6 is identified to be 1,3-butadiene, so we assume that iC_4H_7 can be isomerized to $saxC_4H_7$ (R29). The reaction (R29) with fitted rate constants helps to predict the mole fraction profile of $1,3\text{-}C_4H_6$ in good agreement. Figure 7c shows that the reaction sequence $iC_4H_8 \rightarrow iC_4H_7 \rightarrow saxC_4H_7 \rightarrow 1,3\text{-}C_4H_6$ is very important to the formation of $1,3\text{-}C_4H_6$. Figure 9 shows sensitivity analysis of $1,3\text{-}C_4H_6$ in the pyrolysis of *i*-butene at 1800 K, which confirms the importance of the above reaction sequence. To enhance the knowledge on the mechanism of *i*-butene, detailed theoretical study on the reaction (R29) is desired.

$$iC_4H_7 = saxC_4H_7 \tag{R29}$$

CH_3CCH_2 radical is produced by the C—C scission of iC_4H_8. The sub-mechanism of CH_3CCH_2 was taken from Miller's calculations [42]. CH_3CCH_2 is mostly converted to pC_3H_4 (81%), while the reaction to produce aC_3H_4 only has minor contribution (7%). Besides, 10% CH_3CCH_2 is isomerized to aC_3H_5 by the H attack. It is concluded that *i*-butene is consumed mainly through the reaction sequence $iC_4H_8 \rightarrow iC_4H_7 \rightarrow aC_3H_4 \rightarrow pC_3H_4 \rightarrow C_2H_2$.

4.4. Formation of 1,3-cyclopentadiene and benzene

The formation of the first benzenoid ring in the combustion of nonaromatic fuels is considered to be the rate-controlled step in the formation of polycyclic aromatic hydrocarbons (PAHs) and soot. Two cyclic species including 1,3-cyclopentadiene (C_5H_6) and benzene (C_6H_6) were detected in the pyrolysis of three C_4H_8 isomers. Figures 4f, 5f and 6f show that the predicted mole fractions of C_5H_6 are within the experimental uncertainty. The reaction flux analysis shows that C_5H_6 is mainly produced through the reaction sequence $aC_3H_5 + C_2H_3 \rightarrow lC_5H_7 \rightarrow C_5H_6$ and $iC_4H_5 + CH_3 \rightarrow lC_5H_7 \rightarrow C_5H_6$ in the pyrolysis of three C_4H_8 isomers. In $1\text{-}C_4H_8$ pyrolysis the former pathway is more important, but the second pathway is dominated in both $2\text{-}C_4H_8$ and iC_4H_8 pyrolysis. And C_5H_6 is consumed by the H attack to form C_5H_5, which decomposes rapidly to yield C_2H_2 and C_3H_3 via the ring-opening reaction.

C2 + C4 and C3 + C3 reactions have been included in the present model to predict the formation of C_6H_6. The reaction flux analysis shows that $C_3H_3 + C_3H_3 = C_6H_6$ is the most important pathway for the formation of C_6H_6 in the pyrolysis of three C_4H_8 isomers at 1800 K. Other C3 + C3 reactions such as $aC_3H_4 + C_3H_3 = C_6H_6 + H$ and $pC_3H_4 + C_3H_3 = C_6H_6 + H$ also contribute 20–35% to the formation of C_6H_6. The contribution of $C_4H_4 + C_2H_3 = C_6H_6 + H$ is 20% in the pyrolysis of 2-butene, but quite low in the pyrolysis of 1-butene (4%) and *i*-butene (1%).

5. Conclusion

The experimental study of 1-butene, 2-butene, and *i*-butene pyrolysis from 900 to 1900 K at low pressure has been performed with 1D laminar plug flow reactor using SVUV-PIMS. Pyrolysis species, particularly radicals and isomers, have been detected. A detailed kinetic model of three butene isomers consisting of 76 species and 232 elementary reactions was developed to simulate the pyrolysis processes. Two reactions 2-C_4H_8 = aC_3H_5 + CH_3 and iC_4H_7 = $saxC_4H_7$ are proposed to explain the high concentration aC_3H_5 observed in 2-C_4H_8 pyrolysis and high concentration C_4H_6 observed in iC_4H_8 pyrolysis, respectively. Detailed calculations of the two reactions are desired. Satisfactory agreement is achieved between the experimental and predicted mole fraction profiles of pyrolysis species. Reaction flux analysis indicates that the main decomposition reaction sequences of three butene isomers are 1-butene → aC_3H_5 → aC_3H_4 → pC_3H_4 → C_2H_2, 2-butene → $saxC_4H_7$ → 1,3-C_4H_6 → C_2H_3 → C_2H_2, and iC_4H_8 → iC_4H_7 → aC_3H_4 → pC_3H_4 → C_2H_2.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2011.09.005.

References

[1] A. Chakir, M. Cathonnet, J.C. Boettner, F. Gaillard, Proc. Combust. Inst. 22 (1989) 873–881.
[2] W.J. Pitz, C.K. Westbrook, W.R. Leppard, SAE paper no 912315, 1991.
[3] B. Heyberger, N. Belmekki, V. Conraud, P.A. Glaude, R. Fournet, F. Battin-Leclerc, Int. J. Chem. Kinet. 34 (2002) 666–677.
[4] J.N. Bradley, K.O. West, J. Chem. Soc. Faraday Trans. 1: Phys. Chem. Condens. Phases 72 (1976) 558–567.
[5] S. Santhanam, J.H. Kiefer, R.S. Tranter, N.K. Srinivasan, Int. J. Chem. Kinet. 35 (2003) 381–390.
[6] K. Yasunaga, Y. Kuraguchi, R. Ikeuchi, H. Masaoka, O. Takahashi, T. Koike, Y. Hidaka, Proc. Combust. Inst. 32 (2009) 453–460.
[7] K. Brezinsky, F.L. Dryer, Combust. Sci. Technol. 45 (1986) 225–232.
[8] W. Tsang, J.A. Walker, Proc. Combust. Inst. 22 (1989) 1015–1022.
[9] H.J. Curran, M.P. Dunphy, J.M. Simmie, C.K. Westbrook, W.J. Pitz, Proc. Combust. Inst. 24 (1992) 769–776.
[10] J.C. Bauge, F. Battin-Leclerc, F. Baronnet, Int. J. Chem. Kinet. 30 (1998) 629–640.
[11] P. Dagaut, M. Cathonnet, Combust. Sci. Technol. 137 (1998) 237–275.
[12] V. Dias, J. Vandooren, Fuel 89 (2010) 2633–2639.
[13] N.M. Marinov, W.J. Pitz, C.K. Westbrook, A.M. Vincitore, M.J. Castaldi, S.M. Senkan, C.F. Melius, Combust. Flame 114 (1998) 192–213.
[14] Y.Y. Li, L.D. Zhang, Z.Y. Tian, T. Yuan, J. Wang, B. Yang, F. Qi, Energy Fuels 23 (2009) 1473–1485.
[15] Y.Y. Li, F. Qi, Acc. Chem. Res. 43 (2010) 68–78.
[16] T.C. Zhang, J. Wang, T. Yuan, X. Hong, L.D. Zhang, F. Qi, J. Phys. Chem. A 112 (2008) 10487–10494.
[17] T.C. Zhang, L.D. Zhang, X. Hong, K.W. Zhang, F. Qi, C.K. Law, T.H. Ye, P.H. Zhao, Y.L. Chen, Combust. Flame 156 (2009) 2071–2083.
[18] C.Q. Huang, B. Yang, R. Yang, J. Wang, L.X. Wei, X.B. Shan, L.S. Sheng, Y.W. Zhang, F. Qi, Rev. Sci. Instrum. 76 (2005) 126108.
[19] CHEMKIN-PRO 15092, Reaction Design, San Diego, 2009.
[20] H. Wang, X. You, A.V. Joshi, S.G. Davis, A. Laskin, F. Egolfopoulos, C.K. Law, High-Temperature Combustion Reaction Model of H2/CO/C1-C4 Compounds, 2007. http://ignis.usc.edu/USC_Mech_II.htm>.
[21] S.G. Davis, A.V. Joshi, H. Wang, F. Egolfopoulos, Proc. Combust. Inst. 30 (2005) 1283–1292.
[22] H. Wang, Int. J. Chem. Kinet. 33 (2001) 698–706.
[23] A. Laskin, H. Wang, C.K. Law, Int. J. Chem. Kinet. 32 (2000) 589–614.
[24] A. Laskin, H. Wang, Chem. Phys. Lett. 303 (1999) 43–49.
[25] S.G. Davis, C.K. Law, H. Wang, Combust. Flame 119 (1999) 375–399.
[26] A. Burcat, B. Ruscic, Third Millennium Ideal Gas and Condensed Phase Thermochemical Database for Combustion with Updates from Active Thermochemical Tables, Report TAE960, 2005.
[27] M. Ribaucour, R. Minetti, L.R. Sochet, in: A.R. Burgess, F.L. Dryer (Eds.), Twenty-Seventh Symposium, Combustion Institute, Pittsburgh, 1998, p. 345–351.
[28] R. Minetti, A. Roubaud, E. Therssen, M. Ribaucour, L.R. Sochet, Combust. Flame 118 (1999) 213–220.
[29] S. Touchard, F. Buda, G. Dayma, P.A. Glaude, R. Fournet, F. Battin-Leclerc, Int. J. Chem. Kinet. 37 (2005) 451–463.
[30] M. Yahyaoui, N. Djebaïli-Chaumeix, C.E. Paillard, S. Touchard, R. Fournet, P.A. Glaude, F. Battin-Leclerc, Proc. Combust. Inst. 30 (2005) 1137–1145.
[31] M. Yahyaoui, N. Djebaili-Chaumeix, P. Dagaut, C.E. Paillard, S. Gail, Combust. Flame 147 (2006) 67–78.
[32] M. Mehl, G. Vanhove, W.J. Pitz, E. Ranzi, Combust. Flame 155 (2008) 756–772.
[33] J.H. Kiefer, K.S. Gupte, L.B. Harding, S.J. Klippenstein, J. Phys. Chem. A 113 (2009) 13570–13583.
[34] N. Hansen, W. Li, M.E. Law, T. Kasper, P.R. Westmoreland, B. Yang, T.A. Cool, A. Lucassen, Phys. Chem. Chem. Phys. 12 (2010) 12112–12122.
[35] S. Peukert, C. Naumann, M. Braun-Unkhoff, U. Riedel, Int. J. Chem. Kinet. 43 (2011) 107–119.
[36] W. Tsang, J. Phys. Chem. Ref. Data 20 (1991) 221–273.
[37] W. Tsang, R.F. Hampson, J. Phys. Chem. Ref. Data 15 (1986) 1087.
[38] A. Miyoshi, Int. J. Chem. Kinet. 42 (2010) 273–288.
[39] H.A. Gueniche, P.A. Glaude, R. Fournet, F. Battin-Leclerc, Combust. Flame 151 (2007) 245–261.
[40] R.X. Fernandes, B.R. Giri, H. Hippler, C. Kachiani, F. Striebel, J. Phys. Chem. A 109 (2005) 1063–1070.
[41] H.A. Gueniche, P.A. Glaude, G. Dayma, R. Fournet, F. Battin-Leclerc, Combust. Flame 146 (2006) 620–634.
[42] J.A. Miller, J.P. Senosiain, S.J. Klippenstein, Y. Georgievskii, J. Phys. Chem. A 112 (2008) 9429–9438.
[43] N. Hansen, J.A. Miller, P.R. Westmoreland, T. Kasper, K. Kohse-Höinghaus, J. Wang, T.A. Cool, Combust. Flame 156 (2009) 2153–2164.
[44] S.G. Davis, C.K. Law, H. Wang, J. Phys. Chem. A 103 (1999) 5889–5899.
[45] C.M. Rosado-Reyes, J.A. Manion, W. Tsang, J. Phys. Chem. A 114 (2011) 5710–5717.
[46] J.A. Miller, S.J. Klippenstein, J. Phys. Chem. A 107 (2003) 2680–2692.
[47] L.D. Zhang, J.H. Cai, T.C. Zhang, F. Qi, Combust. Flame 157 (2010) 1686–1697.
[48] Y.Y. Li, J.H. Cai, L.D. Zhang, T. Yuan, K.W. Zhang, F. Qi, Proc. Combust. Inst. 33 (2011) 593–600.
[49] Y.Y. Li, J.H. Cai, L.D. Zhang, J.Z. Yang, Z.D. Wang, F. Qi, Proc. Combust. Inst. 33 (2011) 617–624.
[50] H. Richter, S. Granata, W.H. Green, J.B. Howard, Proc. Combust. Inst. 30 (2004) 1397–1405.
[51] G.B. Bacskay, J.C. Mackie, Phys. Chem. Chem. Phys. 3 (2001) 2467–2473.
[52] H. Wang, E. Dames, B. Sirjean, D.A. Sheen, R. Tangko, A. Violi, J.Y. W. Lai, F.N. Egolfopoulos, D.F. Davidson, R.K. Hanson, C.T. Bowman, C.K. Law, W. Tsang, N.P. Cernansky, D.L. Miller, R.P. Lindstedt, JetSurF version 2.0 September 19, 2010. <http://melchior.usc.edu/JetSurF/JetSurF2.0>.
[53] G. Blanquart, P. Pepiot-Desjardins, H. Pitsch, Combust. Flame 156 (2009) 588–607.
[54] V. Detilleux, J. Vandooren, J. Phys. Chem. A 113 (2009) 10913–10922.
[55] R. Sivaramakrishnan, K. Brezinsky, H. Vasudevan, R.S. Tranter, Combust. Sci. Technol. 178 (2006) 285–305.
[56] A. D'Anna, A. D'Alessio, J. Kent, Combust. Flame 125 (2001) 1196–1206.
[57] H.Y. Lee, V.V. Kislov, S.H. Lin, A.M. Mebel, D.M. Neumark, Chem. Eur. J. 9 (2003) 726–740.